Virtual GlobeTrotter :
Travel and see nice spots

*** even when you have to stay home**

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Team Role Distribution

Viola

Scrum Master
API
UI
Trello
Connectivity

Atvars

Development
AI
Unity
Connectivity

Isabelle

Product Owner
Database
Documentation
Connectivity

Project Description

What? 360° AI generated immersion based on voice input to simulate a study trip.

The "Virtual Globetrotter" project aims to provide children with an opportunity to experience the world virtually using our immersive room. Children can explore locations using an interactive map. These virtual trips will utilize AI-generated images, speech recognition, and real-time audio to create an immersive and enriching experience.

For? Children around 10 yrs?

Children who are vulnerable after surgery, during a treatment or because of a disability are not able to go on the trips the school or their family is doing. The child can go into the immersive room and travel to several spots in the world!

Where? Immersive Room in Forum Core Howest

The child is in the immersive room and has a tablet or phone with the user interface connected to the pc in the immersive room. The child chooses a continent shown on the user interface and a 360° image is shown and music is played corresponding to the choice.

The child is then asked to say a location, landmark or point of interest it wants to explore. For example, the beach, the pyramids or Disneyland.

The voice input is used by an AI model to generate another 360° image to show in the immersive room.

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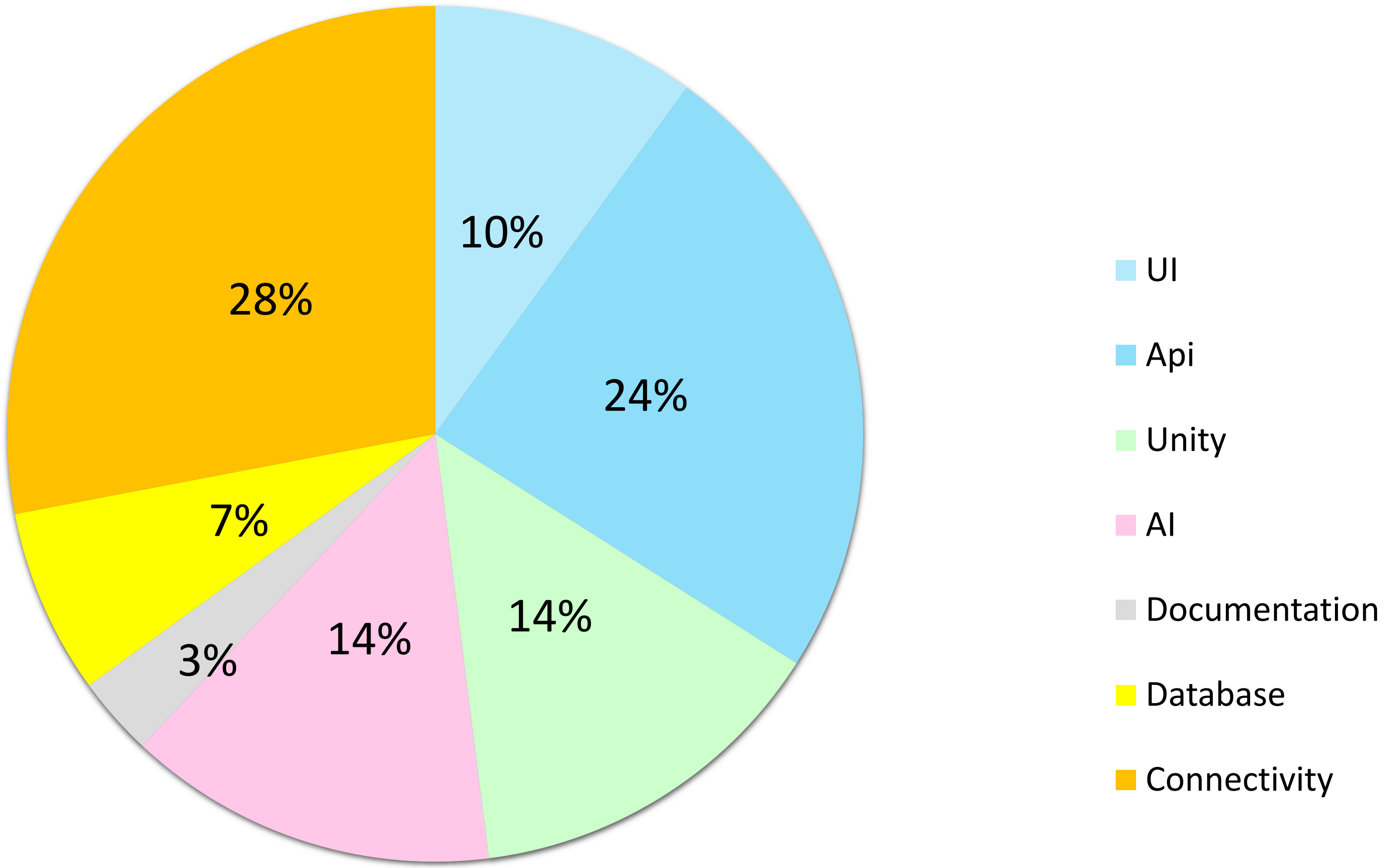
Story points



Trello

•create a board and share it with the Scrum team & coaches

•set up the product backlog

•define sprints

•add tasks and assign deadlines

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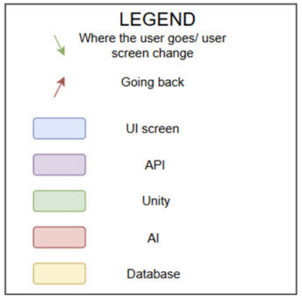
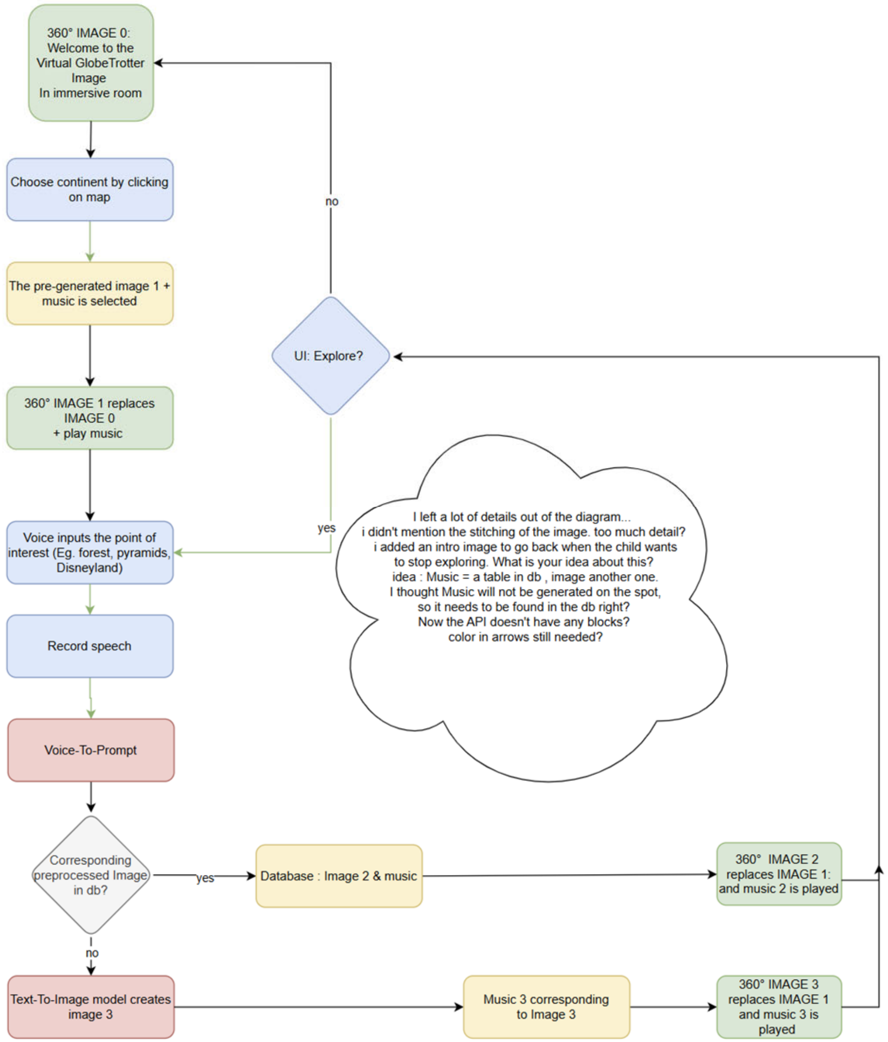
Technical solution Diagram

To update Thursday 07/11





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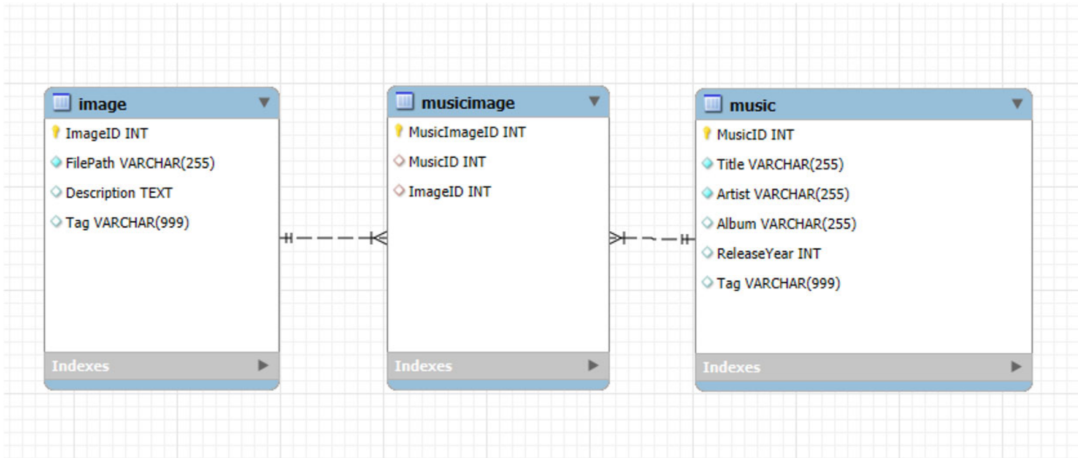
Technical solution UI: You decide!

User Interface: The focus will be on creating a user-friendly interface that is accessible and enjoyable for children of different ages and abilities.

Technical solution API: help us

- Dynamic Environment Adaptation via Speech Input: The environments will dynamically adapt based on the child's spoken input. Speech-to-text technology combined with prompt engineering will ensure that elements in the environment change according to what the child says, making the experience more engaging and interactive.

Technical solution Database: find the best one!



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Technical solution AI: or Else Create It!

AI-Generated Environments: The chosen locations will be brought to life with AI-generated 360-degree images and videos, offering immersive experiences such as nature (forests, snow-covered landscapes) or cities (Paris, Amsterdam, etc.).

No real time music generation?

Real-time Music Generation: Background music will be generated in real-time and tailored to match the location and the interactions of the child, adding to the immersive atmosphere.

Music? Copyrights?

Technical solution Unity: Eyes and Ears Ready?

Immersive Room:

• Mobile immersive room (Dome) with 360° projection and audio setup

• Dimensions: 5x5 meters with four large white screens, four 8000-lumen projectors, and sound boxes in each corner. Hardware is positioned next to the immersive room.

• VIVE trackers for motion tracking

• Compatible with Unity, Unreal Engine, and 360-degree videos

Technical solution Connectivity

Connecting everything to 1 experience